Data Description

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Objectives

- Summarize data using the measures of central tendency, such as the mean, median, mode, and midrange.
- Describe data using the measures of variation, such as the range, variance, and standard deviation.
- Identify the position of a data value in a data set using various measures of position, such as z-scores and percentiles.

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Measures of Central Tendency

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Measures of Central Tendency

A measure of central tendency is a single value that attempts to describe a set of data by identifying the central position within that set of data.

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Measures of Central Tendency

- A **statistic** is a characteristic or measure obtained by using the data values from a sample.
- A **parameter** is a characteristic or measure obtained by using the data values from a specific population.

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Mean (arithmetic average)

- The **mean** is defined to be the sum of the data values divided by the total number of values.
- We will compute two means: one for the sample and one for a finite population of values.
- The mean, in most cases, is not an actual data value.

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Sample Mean

The symbol $\overline{X}$ represents the sample mean.

$\overline{X}$ is read as "X-bar". The Greek symbol $\sum$ is read as "sigma" and it means "to sum".

$$\overline{X} = \frac{X_1 + X_2 + \ldots + X_n}{n}$$

$$= \frac{\sum X}{n}.$$

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Sample Mean - Example

The ages in weeks of a random sample of six kittens at an animal shelter are 3, 8, 5, 12, 14 and 12. Find the average age of this sample.

Solution: $\bar{x}=(3+8+5+12+14+12)/6 = 9$

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Population Mean

The Greek symbol μ represents the population mean. The symbol μ is read as "mu".

N is the size of the finite population.

$$\mu = \frac{X_1 + X_2 + \ldots + X_N}{N}$$

$$= \frac{\sum X}{N}.$$

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Population Mean - Example

A small company consists of the owner, the manager, the salesperson, and two technicians. The salaries are listed as $50 000, 20 000, 12 000, 9 000 and 9 000 respectively. Find the population mean. (Assume this is the population)

Solution: $\mu=(3+8+5+12+14+12)/6 = 9$

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Weighted Mean

- The **weighted mean** is used when the values in a data set are not all equally represented.
- The **weighted mean of a variable *X*** is found by multiplying each value by its corresponding weight and dividing the sum of the products by the sum of the weights.

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Weighted Mean

The weighted mean

$$\overline{X} = \frac{w_1 X_1 + w_2 X_2 + \ldots + w_n X_n}{w_1 + w_2 + \ldots + w_n} = \frac{\sum wX}{\sum w}$$

where $w_1, w_2, \ldots, w_n$ *are the weights for the values* $X_1, X_2, \ldots, X_n$.

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Weighted Mean
-Example

- A student is enrolled in a biology course where the final grade is determined based on the following categories: tests 40%, final exam 25%, quizzes 25%, and homework 10%. The student has earned the following scores for each category: tests-83, final exam-75, quizzes-90, homework-100. We need to calculate the student's overall grade.

- **Solution:**
$\bar{x}$=(0.40(83)+0.25(75)+0.25(90)+0.10(100)) / (0.40+0.25+0.25+0.10) = **84.45** or **84%**

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Weighted Mean
-Example

Fat Grams Using the weighted mean, find the average number of grams of fat per ounce of meat or fish that a person would consume over a 5-day period if he ate these:

Meat or fish	Fat (g/oz)
3 oz fried shrimp	3.33
3 oz veal cutlet (broiled)	3.00
2 oz roast beef (lean)	2.50
2.5 oz fried chicken drumstick	4.40
4 oz tuna (canned in oil)	1.75

Source: *The World Almanac and Book of Facts.*

- **Solution:**

$\bar{x}=(3(3.33)+3(3.00)+2(2.50)+2.5(4.40)+4(1.75)) / (3+3+2+2.5+4) =$ **2.90**

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Median

- When a data set is ordered, it is called a **data array**.
- The **median** is defined to be the midpoint of the data array.
- The symbol used to denote the median is **MD**.

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Median - Example

- The weights (in pounds) of seven army recruits are 180, 201, 220, 191, 219, 209, and 186. Find the median.

- **Solution:**
- **Data array:** 180, 186, 191, 201, 209, 219, 220.
- The median, MD = 201.

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Median

- In the previous example, there was an odd number of values in the data set. In this case it is easy to select the middle number in the data array.
- When there is an even number of values in the data set, the median is obtained by taking the average of the two middle numbers.

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Median - Example

- The ages of 10 college students are: 18, 24, 20, 35, 19, 23, 26, 23, 19, 20. Find the median.

- **Solution:**
- **Data array:** 18, 19, 19, 20, 20, 23, 23, 24, 26, 35.
- The median, MD = (20 + 23)/2 = 21.5.

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Mode

- The **mode** is defined to be the value that occurs most often in a data set.
- A data set can have more than one mode.
- A data set is said to have **no mode** if all values occur with equal frequency.

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Mode - Examples

- The following data represent the duration (in days) of U.S. space shuttle voyages for the years 1992-94. Find the mode.
- Data set: 8, 9, 9, 14, 8, 8, 10, 7, 6, 9, 7, 8, 10, 14, 11, 8, 14, 11.
- **Solution:**
- Ordered set: 6, 7, 7, 8, 8, 8, 8, 8, 9, 9, 9, 10, 10, 11, 11, 14, 14, 14.
- **Mode = 8.**

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Mode - Examples

- Six strains of bacteria were tested to see how long they could remain alive outside their normal environment. The time, in minutes, is given below. Find the mode.
- Data set: 2, 2, 3, 3, 5, 5, 7, 7, 8, 8, 10, 10.

- **Solution:**
- There is **no mode** since each data value occurs equally with a frequency of two.

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Mode - Examples

- Eleven different automobiles were tested at a speed of 15 mph for stopping distances. The distance, in feet, is given below. Find the mode.
- Data set: 15, 18, 18, 18, 20, 22, 24, 24, 24, 26, 26.

- **Solution:**
- There are two modes (bimodal). The values are 18 and 24.

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Midrange

- The midrange is found by adding the lowest and highest values in the data set and dividing by 2.
- The symbol that is used to represent the midrange is MR.

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Midrange - Example

- Last winter, the city of Brownsville, Minnesota, reported the following number of water-line breaks per month. The data is as follows: 2, 3, 6, 8, 4, 1. Find the midrange.
- **Solution:**

$$MR = (1 + 8)/2 = 4.5.$$

- ***Note:*** Extreme values influence the midrange and thus may not be a typical description of the middle.

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Distribution Shapes

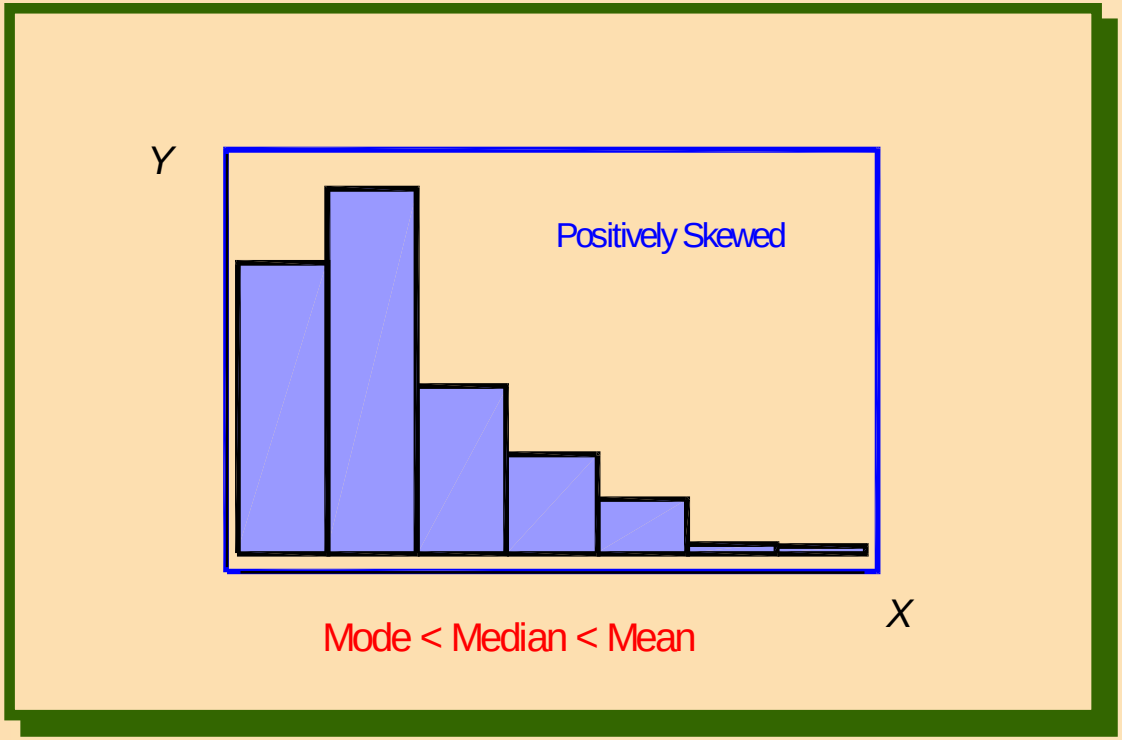
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Distribution Shapes

- Frequency distributions can assume many shapes.
- The three most important shapes are positively skewed, symmetrical, and negatively skewed.

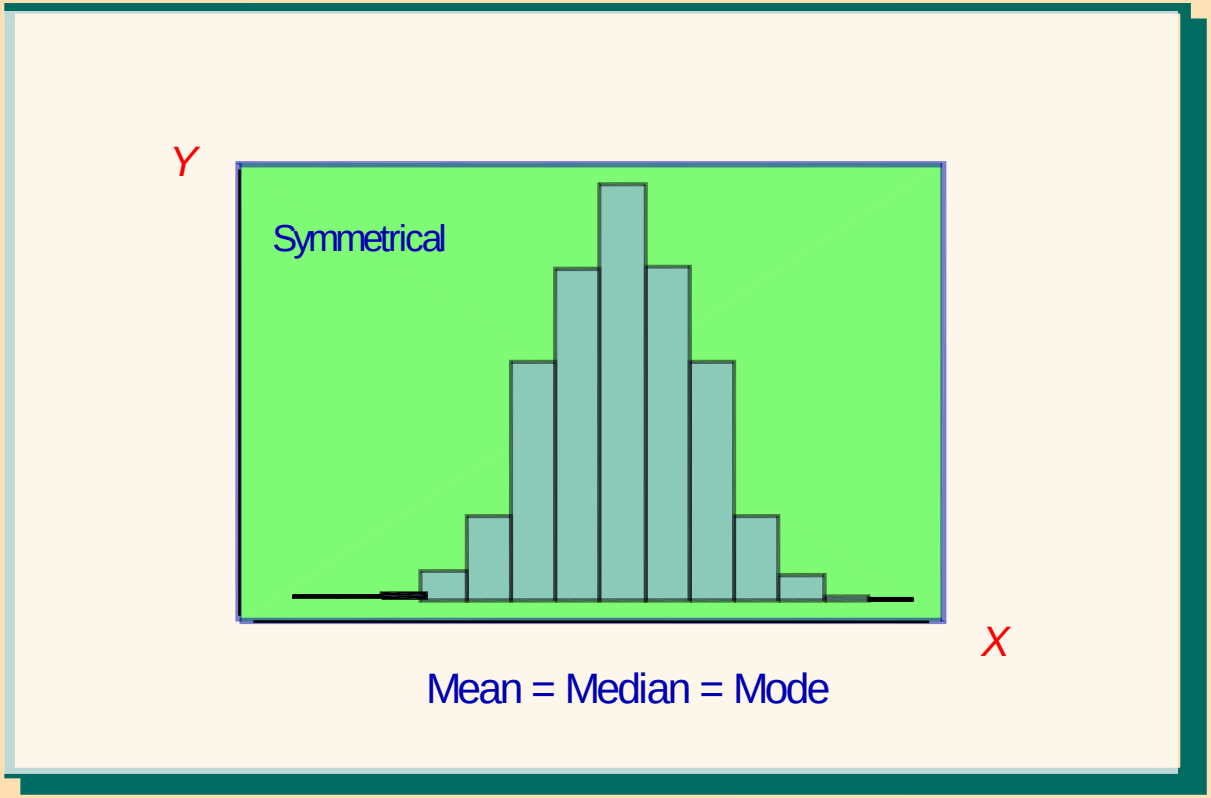
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Positively Skewed



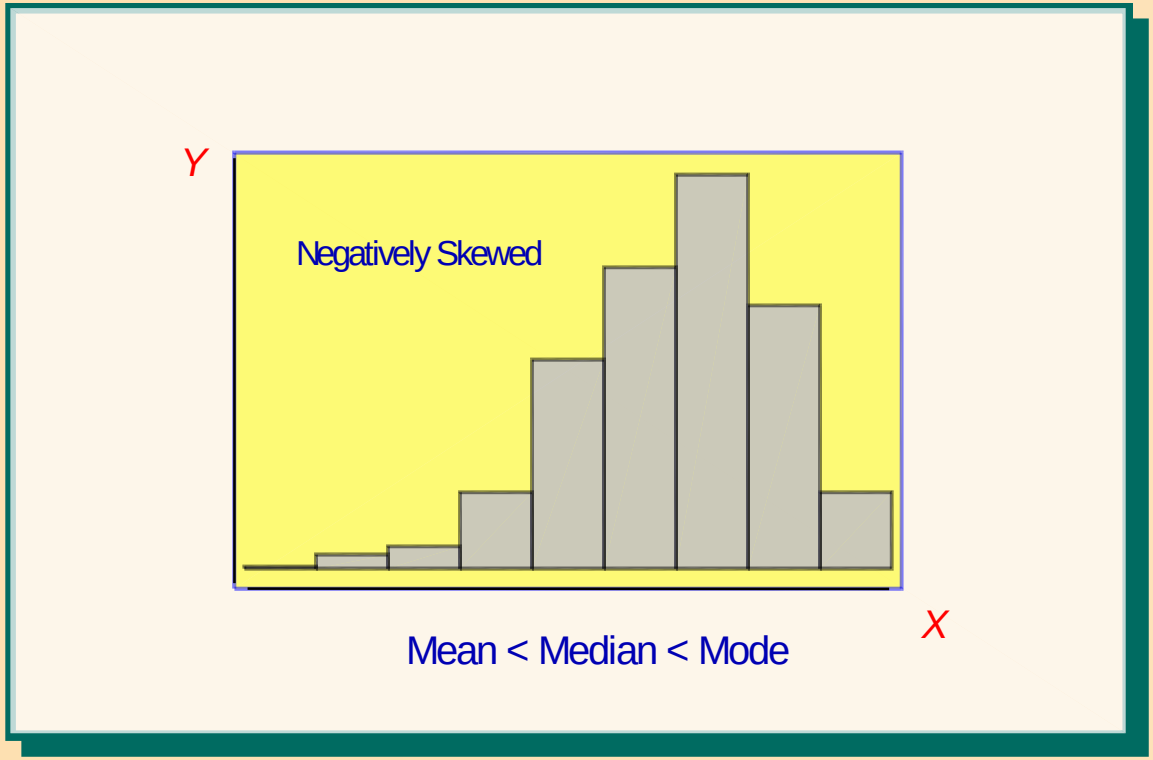
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Symmetrical



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Negatively Skewed



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When to use the mean, median and mode?

Type of Variable	Best measure of central tendency
Nominal	Mode
Ordinal	Median
Interval/Ratio (not skewed)	Mean
Interval/Ratio (skewed)	Median

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Measures of Variation

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Measures of Variation

- **Measure of Variation**

-measures how scattered the data values in a sample or population.

-also called a measure of dispersion.

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Measures of Variation - Range

- The **range** is defined to be the highest value minus the lowest value. The symbol ***R*** is used for the range.
- ***R*** = highest value – lowest value.
- Extremely large or extremely small data values can drastically affect the range.

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Measures of Variation – Population Variance

The variance is the average of the squares of the distance each value is from the mean.

The symbol for the population variance is σ^2 *(*σ *is the Greek lowercase letter sigma)*

$$\sigma^2 = \frac{\sum (X - \mu)^2}{N}, \quad \text{where}$$

X = *individual value*

μ = *population mean*

N = *population size*

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Measures of Variation - Population Standard Deviation

The standard deviation is the square root of the variance.

$$\sigma = \sqrt{\sigma^2} = \sqrt{\frac{\sum (X - \mu)^2}{N}}.$$

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Measures of Variation - Example

- Consider the following data to constitute the population: 10, 60, 50, 30, 40, 20. Find the variance.

- **Solution:**
- The mean μ = (10 + 60 + 50 + 30 + 40 + 20)/6 = 210/6 = 35.
- The variance σ^2 = $((10-35)^2 + (60-35)^2 + (50-35)^2 + (30-35)^2 + (40-35)^2 + (20-35)^2)/6$ = 291.67.

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Measures of Variation - Sample Variance

The unbiased estimator of the population variance or the sample variance is a statistic whose value approximates the expected value of a population variance. It is denoted by s^2, *where*

$$s^2 = \frac{\sum (X - \bar{X})^2}{n-1}, \text{ and}$$

$\bar{X}$ = *sample mean*

n = *sample size*

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Measures of Variation - Sample Standard Deviation

The sample standard deviation is the square root of the sample variance.

$$s = \sqrt{s^2} = \sqrt{\frac{\Sigma (X - \bar{X})^2}{n-1}}.$$

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Sample Variance - Example

- **Find the variance and standard deviation for the following sample: 16, 19, 15, 15, 14.**
- **Solution:**
- **The sample mean ($\bar{x}$) = (16 + 19 + 15 + 15 + 14)/5 = 15.8.**
- **The sample variance (s^2) = ((16-15.8)^2 + (19-15.8)^2 + (15-15.8)^2 + (15-15.8)^2 +(14-15.8)^2)/(5-1) = 3.7**
- **The sample SD (s) = $\sqrt{3.7}$ = 1.92**

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Measures of Position

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Measures of Position
- z score

- The standard score or *z* score for a value is obtained by subtracting the mean from the value and dividing the result by the standard deviation.
- The symbol *z* is used for the *z* score.

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Measures of Position - z-score

- **The *z* score represents the number of standard deviations a data value falls above or below the mean.**

For samples:

$$z = \frac{X - \overline{X}}{s}.$$

For populations:

$$z = \frac{X - \mu}{\sigma}.$$

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Measures of Position - z-score Example

- A student scored 65 on a statistics exam that had a population mean of 50 and a population standard deviation of 10. Compute the *z*-score.

- **Solution:**
- $z = (65 - 50)/10 = 1.5$.
- That is, the score of 65 is 1.5 standard deviations **above** the mean.
- **Above -** since the *z*-score is positive.

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Measures of Position
- z-score Example

A student scored 65 on a Calculus test that had a sample mean of 50 and a sample standard deviation of 10; she scored 30 on a History test with a sample mean of 25 and a sample standard deviation of 5. Compare her relative positions on the two tests.

First, find the z scores. For calculus the z score is

$$z = \frac{X - \overline{X}}{s} = \frac{65 - 50}{10} = 1.5$$

For history the z score is

$$z = \frac{30 - 25}{5} = 1.0$$

Since the z score for calculus is larger, her relative position in the calculus class is higher than her relative position in the history class.

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Measures of Position - Percentiles

- Percentiles divide the distribution into 100 groups.
- The P_k percentile is defined to be that numerical value such that at most k% of the values are smaller than P_k and at most (100 – k)% are larger than P_k in an ordered data set.

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Measures of Position - Percentiles

- **The percentile corresponding to a given value (*X*) is computed by using the formula:**

$$Percentile = \frac{number\ of\ values\ below\ X + 0.5}{total\ number\ of\ values} \cdot 100$$

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Measures of Position – Percentiles Example

- A teacher gives a 20-point test to 10 students. Find the percentile rank of a score of 12. Scores: 18, 15, 12, 6, 8, 2, 3, 5, 20, 10.

- Solution:
- Ordered set: 2, 3, 5, 6, 8, 10, 12, 15, 18, 20.
- Percentile = [(6 + 0.5)/10](100%) = 65th percentile.
- Student did better than 65% of the class.

Reference:

Bluman, A. (2012). *Elementary Statistics: A Step by Step Approach*,8e. McGraw-Hill Higher Ed.

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End of Lecture



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